

Energy-efficient Analog Integrated Circuits

Sensor Interface

Layout Guide and Bonus Exercise

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1. Introduction

The schematic view is useful for understanding a circuit conceptually, but eventually, it needs to be translated into a physical layout. The layout provides a top-down view of the circuit components and uses different colors to identify various layers. The placement of components is the first step, followed by routing to make the necessary connections between them. Routing is achieved by drawing traces in the metal layers, and vias are used to connect different layers. Once the layout is complete, it must be verified to ensure it meets design rules and implements the desired circuit correctly. After successful verification, a simulation can be performed using information extracted from the layout to predict the behavior of the circuit in measurements.

2. Getting Started

We are going to draw the layout of a simple differential pair with active load. You can find the schematic and the final layout in /home/eeaic_30fs23/5T_OTA_LayoutExample.

1. Copy the entire folder in your working library using

```
cp -r /home/eeaic_30fs23/5T_OTA_LayoutExample <your cockpit folder>/dfii/<your library>
```

2. Open the schematic view. It consists of 5 transistors (M0 to M4) that represent a transconductance amplifier and some dummy transistors that are always turned off. Click on Launch > Layout XL. A window will show up. In the “Layout” section, select “Create New”, and in the “Configuration” section, select “Automatic”. Click “OK”. A new window will show up. Make sure that layout is written in the “View” field. Click “OK”. You should see the “Virtuoso Layout Suite XL” window open.

3. The Layout Editor

Before starting the actual layout of our circuit we need to discuss some general information on the layout editor. First, the layers palette, shown to the right, allows you to manage the different layers in your layout view. The top box, currently set to M1 (Metal 1) drawing, shows the active layer. This is the layer that gets drawn when you create an object.

There are four buttons worth noting: “AV”, “NV”, “AS”, and “NS”. “AV” stands for “All Visible” and displays all layers, while “NV” means “None Visible” and hides all layers except the selected one. Once “NV” is clicked, layers can be individually turned on by selecting their checkbox in the “V” column. Similarly, “AS” and “NS” mean “All Selectable” and “None Selectable” respectively, which enable or disable the selection of specific layers. This can also be controlled for each layer by ticking their checkbox in the “S” column. Hiding and showing layers strategically during routing will make your task much easier.

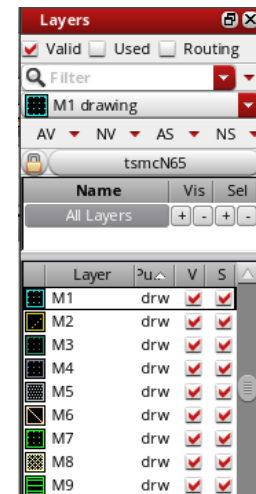


Fig. 1. Layers palette

Also, as you continue working with Virtuoso Layout Editor, you will often find it useful to zoom parts of your layout. You can use the scroll wheel of your mouse, or simply remember these useful hotkeys:

- **Z**: Zoom-in the selected area (hit **Z**, left-click, and define the zoom area)
- **Ctrl + Z**: Zoom-in by 2x
- **Shift + Z**: Zoom-out by 2x
- **F**: Zoom fit

A more detailed list of shortcuts is provided for your reference at the end of this guide.

Finally, press **E** to enter the display options. In the grid-control section, make sure that the X/Y snap spacing is set to 0.005. This defines your layout grid resolution.

4. Layout Example – Differential Pair with Active Load

4.1. Device Placement

You can import your components from the schematic to the layout view by using the button below, or by clicking Connectivity > Generate > All from source.



However, in this case, we are not going to import the components from the schematic, but we will instantiate the transistors directly. Press **I** and select the pch_lvt device from the tsmcN65 library. Select a length L of 500 nm and a width W of 10 μm , with 24 fingers. Place the device by clicking on any location in your layout editor. You should see the transistor in Fig. 2. If you only see a red box representing the transistor outline you can press **Shift + F** and **Ctrl + F** to respectively show or hide lower hierarchy levels. Fig. 2 also shows how we are going to implement the p-channel input pair consisting of transistors M1 and M2. We will make an interdigitated layout where M1 and M2 are

divided into four parts with two fingers each that we will connect in parallel. In the layers palette, click on *Used* and you should see the following layers:

- NW: n-well indicator
- OD: gate oxide and diffusion
- PO: polysilicon
- PP: p+ implant
- M1 (drw): lowest metal level
- CO: contacts from PO/OD to M1

For n-channel devices, you would see the NP layer instead of PP, representing n+ implants. We can now move forward and instantiate also the transistor M0. Again, use the shortcut *I* and select $L = 500\text{ nm}$, $W = 2\mu\text{m}$ and number of fingers = 24. Place this transistor as shown in Fig. 3. Use the command align (*A*) to align

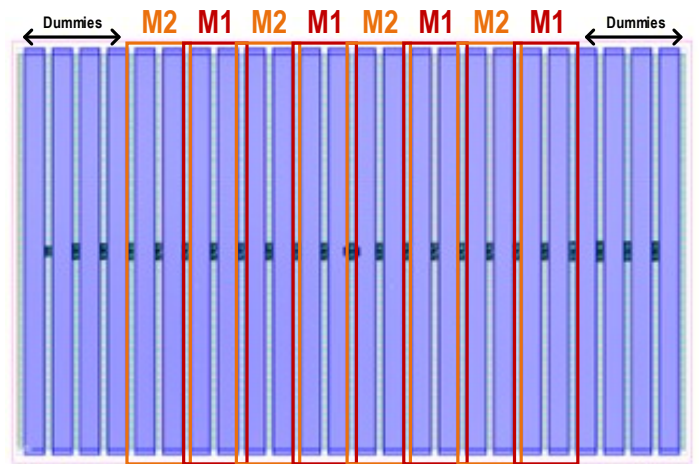


Fig. 2. Input source-coupled pair

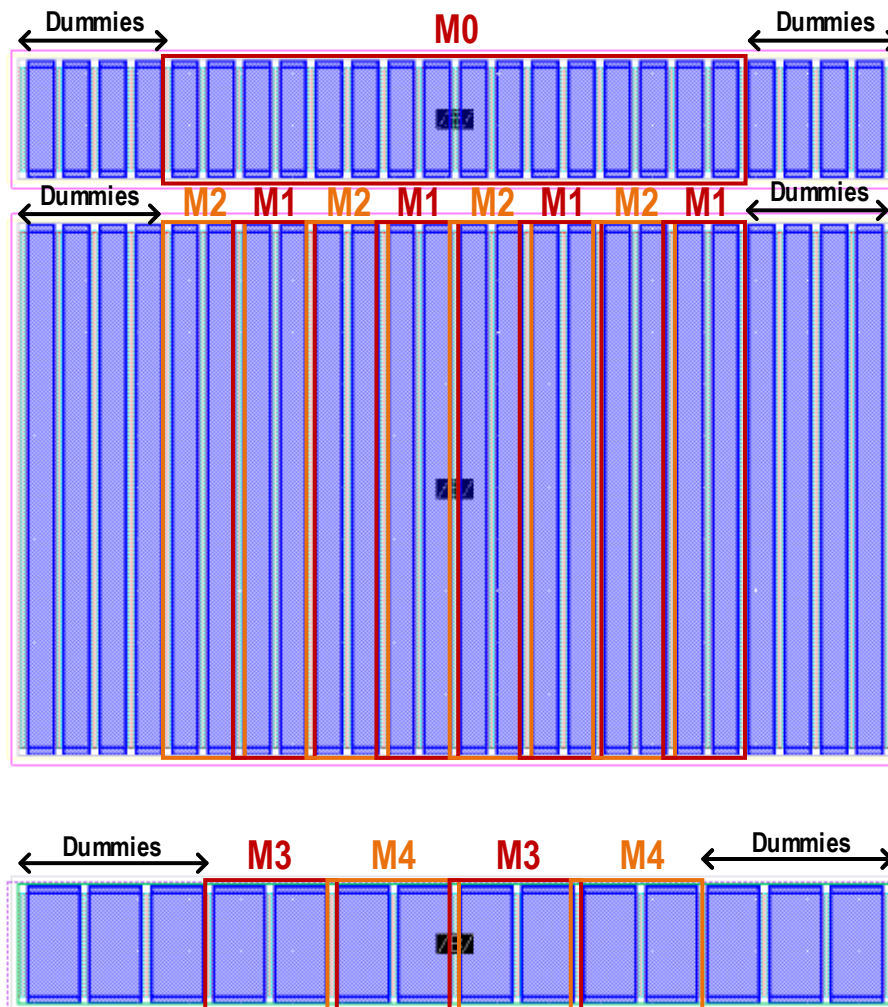


Fig. 3. Device placement

the two transistors. Press **A** on your keyboard, click on any vertical edge in transistor M0, and click on the same edge in the input-pair devices. This will align the two devices along the x-axis. Now click on the transistor M0, press **Q**, and select **Parameter**. Here you have access to the main parameters that characterize the layout of your transistor. There are several options:

- Enabling the **routePolyDir** option will automatically connect the gates of different fingers. This will also allow you to automatically place the gate contacts if you also choose the **polyContacts** option.
- Enabling the **route_Source_Drain** option will automatically connect the source and drain terminals.
- You can also choose to swap the source and drain using the **S D swap** option.

Tick the **S D swap** option and press OK. We will finally place the n-channel active load. Instantiate the `nch_hvt` device and select $L = 1\ \mu\text{m}$, $W = 2\ \mu\text{m}$ and Number of fingers = 14. Again, in this case, four fingers will be used for transistor M3, four fingers will be used for transistor M4, and the remaining fingers are dummies. Now your layout should look similar to Fig. 3. In this case, we will place our p-channel devices in the same n-well. In order to do this, we need to create a rectangle (**R**). First, select the NW layer, then press **R** on your keyboard and draw the n-well for our p-channel devices. We will do the same thing with the PP layer. Refer to Fig. 4 and to the provided layout example to see how the n-well and p+ implant layers are drawn. Finally, we need to create contacts for the n-well and p-substrate. This is needed to connect the bulk of p-channel and n-channel devices to V_{DD} and V_{SS} respectively. We will do this by using two guard rings. A guard ring is an array of substrate contacts that encloses one or more devices and is connected to a clean supply (V_{DD} or V_{SS}). This helps minimize the effect of substrate noise when laying out sensitive analog blocks. In order to place a guard ring, use the guard ring command (**Shift + G**) and use the rectangular **NWellGuardring** template from the

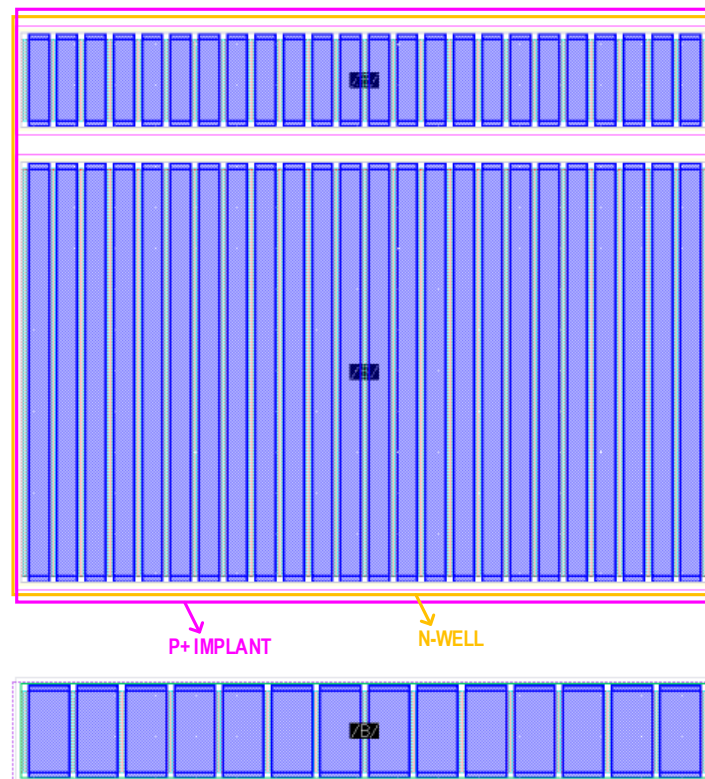


Fig. 4. N-well and p+ implant

tsmcN65 technology library. Make sure to select the rectangular option (rather than rectlinear) in your guard ring window and set the parameter *enclose by* to 0.25 μm (note that all dimensions are usually expressed in μm). Now select your p-channel devices in your layout editor and press enter. You should see the guard ring in your layout. Repeat this operation for your n-channel devices but now select the *PSubGuardring* template.

Finally, we may want to draw an additional p-substrate guard ring around the n-well. You can do this by again using the guard-ring command (*Shift + G*) and selecting the area where you drew the n-well guard ring. This time in order to respect the minimum distance between n+ implants and p+ implants you need to set “*enclose by*” as 0.5 μm . The resulting layout is shown in Fig. 5.

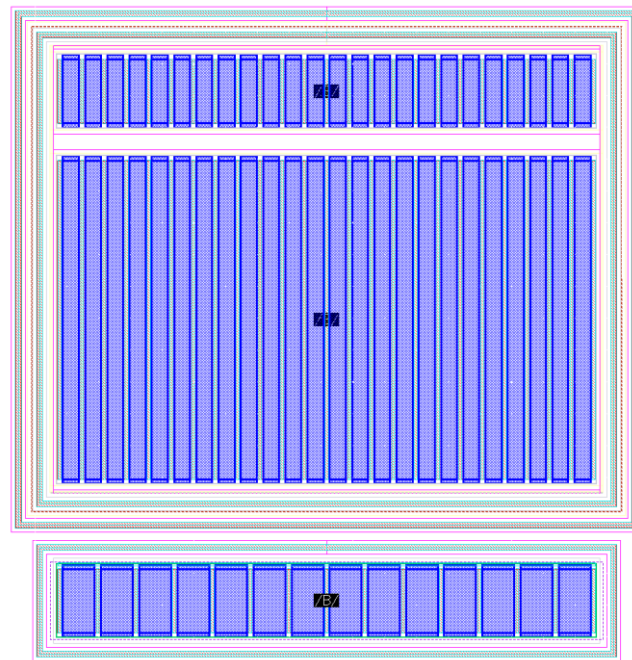


Fig. 5. Guard rings

4.2. Routing

After the devices are placed, we can now start to make all the necessary connections. This is done in the layout by drawing traces in the different metal layers. For the specific process we are using, there

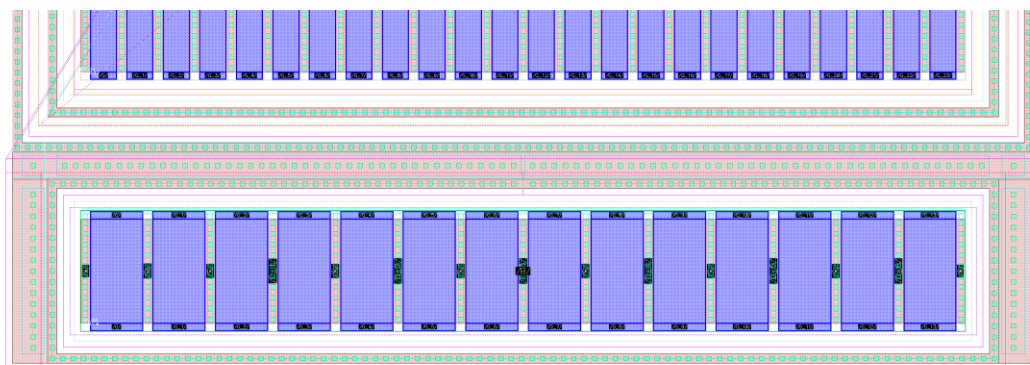


Fig. 6. Connection of p-substrate guard rings

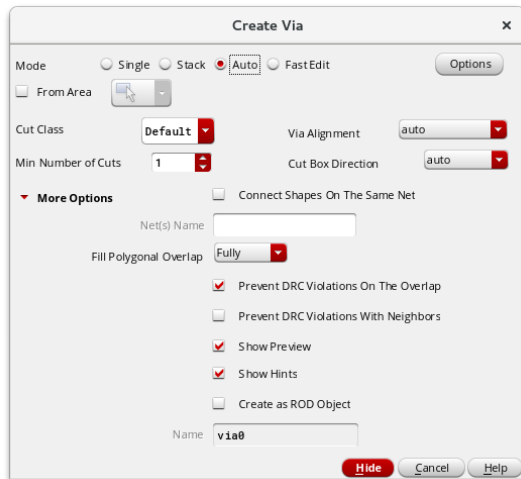


Fig. 7. Via options

this can be done by just creating rectangles to enlarge the guard ring enclosing the n-channel devices. You can merge the two guard rings by just creating some rectangles in the M1 (purpose *drw*), PP (p+ implant) and OD (diffusion) layers. Fig. 6 shows how the two guard rings are connected. Once you have connected the p-substrate guard rings you can create the substrate contacts. In general, connections between different routing layers are done by placing vias and contacts. You can use the shortcut **O** to create such connections. Press **O** and make sure you select **Auto** in the via options window. This will allow you to create vias/contacts between any visible layer by just clicking on the area where you want to place them. Click on any M1 rectangle and you should now see the substrate contacts in the CO layer (Fig. 6).

After the guard rings are connected, we will connect the gate terminals of the different fingers for each transistor. The process is really similar to what we have seen for the p-substrate guard rings, as again we need to create some rectangles in the PO layers to connect the gates and place the gate contacts from the polysilicon (PO) to the lowest metal level (M1). First, draw a rectangle to connect two adjacent fingers, as shown in Fig. 8. Then, we can replicate this rectangle to connect all the fingers representing the transistor M2. Use the ruler (**K**) to measure the distance between one pair of two fingers and the

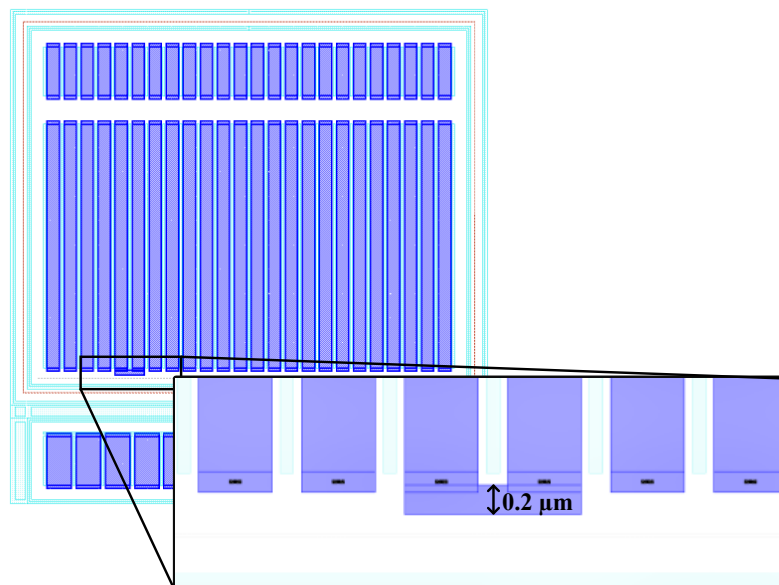


Fig. 8. Gate connection

are 9 metal layers, named M1 through M9. To ensure that they can be properly fabricated, metal traces must also follow certain design rules (e.g., regarding their minimum and maximum width and spacing). For the exact rules, check the tsmcN65 design rule manual (DRM) under docs/T-N65-CL-DR-001 2 4.pdf in your cockpit directory. As you learned before, you can create a rectangle by pressing **R** on your keyboard, then clicking once to define one of the rectangle's corners, and then a second time to define the opposite corner. Alternatively, you can create a path by tapping **P**. Click once every time you want to make a turn in your path, and double-click to end the path.

We can start the routing phase by connecting the two p-substrate guard rings. In principle,

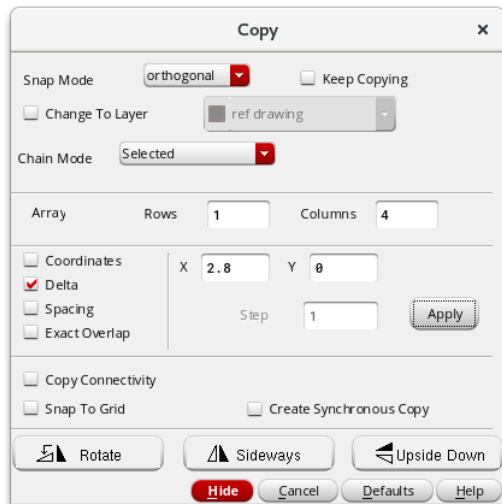


Fig. 9. Copy options

next one. In this case, the distance is $2.8\ \mu\text{m}$. You can remove the ruler by pressing **Shift + K** on your keyboard. Now use the copy command (**C**) and then press **F3**. Usually, when a command is active, **F3** allows you to access several options for that command. You should see the window in Fig. 9. First, we can specify that we want to create 4 copies across a single row (i.e., number of rows = 1 and number of columns = 4). Specify a pitch of $2.8\ \mu\text{m}$ and click on “Apply”. Now use the path command (**P**) to create a trace in metal 1 which we will use to connect the gates. Align the path to the center of the rectangle that you just drew in the PO layer, using the align command (**A**), and create contacts (**O**) from the PO layer to the metal 1. In the same way, you can connect the gate terminals for transistors M0, M1, M3, M4, and the dummy transistors. The resulting layout is shown in Fig. 10. Note that dummy transistors have been connected either to the p-substrate guard ring or n-well guard ring as these nets represent our supplies

(V_{SS} and V_{DD} respectively). This will ensure that these devices are turned off.

At this point, to avoid design rule violations, we need to slightly extend the height of the NP rectangle

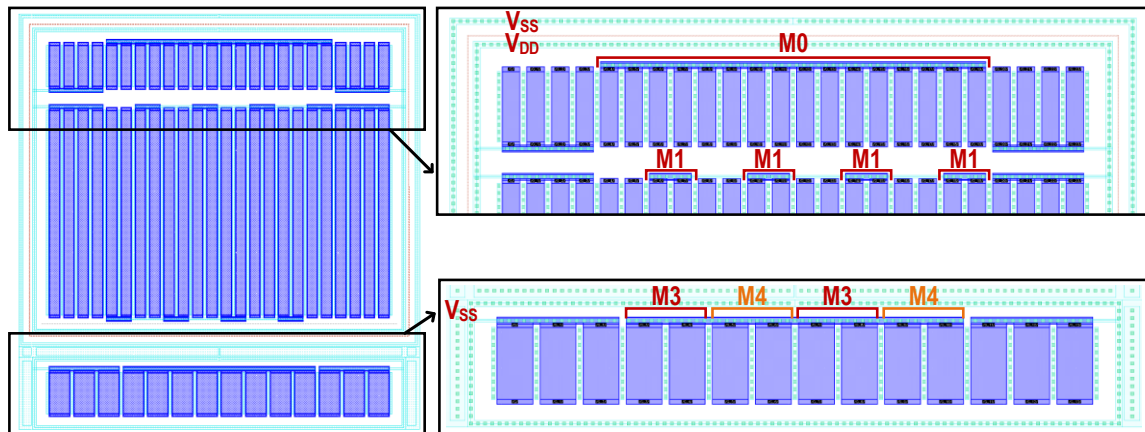


Fig. 10. Gate connections for M0, M2, M4, M3 and the dummy transistors

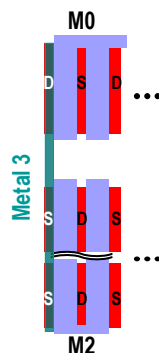


Fig. 11. Connection between the source of transistor M1 and the drain of transistor M0

enclosing our n-channel devices. Any PO object should be enclosed by either NP or PP by at least $0.15\ \mu\text{m}$. This can be accomplished by simply creating an NP rectangle that encloses all our PO rectangles by at least $0.15\ \mu\text{m}$.

We will then connect the source of transistor M0 to the source of transistors M1 and M2. Select layer M3 (purpose draw) in the layers palette and draw a vertical trace (**P**) as shown in Fig. 11. You may need to uncheck **Used** and check **Routing** in your layers palette to be able to see M3. We can then place vias from M3 to M1 by using the Create Via

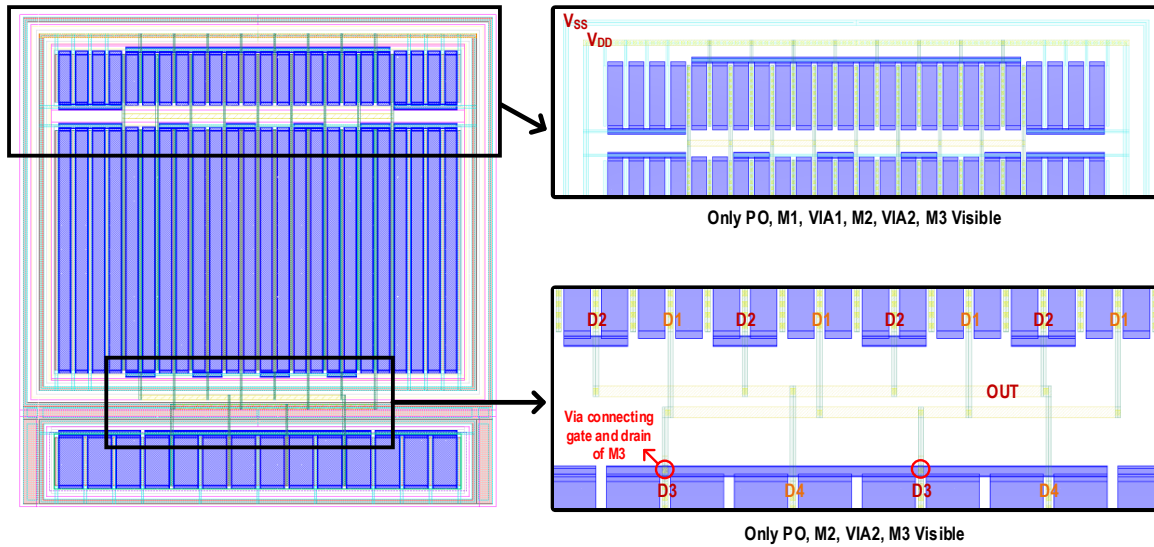


Fig. 12. OTA layout after routing

command (**O**). Now create 9 copies (**C**) with a 1.4 μm pitch and connect them using the M2 (drw) layer as shown in Fig. 12 (on the left).

From now on, you have all the necessary information to complete the routing phase. We have to connect the drains of transistors M1 and M3 as well as the drains of transistors M2 and M4. The gate of transistor M3 is connected to its drain and the source terminals of M3 and M4 should be connected to the V_{SS} guard ring. Similarly, the source of M0 will be connected to the n-well guard ring. Moreover, the source and drain terminals of n-channel dummy transistors will be tied to V_{SS} while they need to be tied to V_{DD} for p-channel devices.

- You can start by connecting the drains of M1/M2 to M3/M4 using the 3rd metal layer vertically and the 2nd metal layer horizontally. Refer to Fig. 12 to see how the drains have been connected.
- Similarly, connect the source of M0 to V_{DD} using Metal 2 and Metal 3.

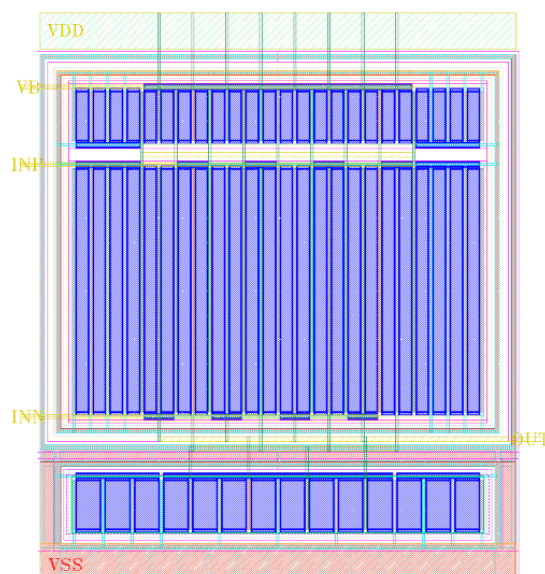


Fig. 13. Final layout

- Then, you can create some vias from Metal 3 to Metal 1 to connect the gate and drain terminals of transistor M3.
- Finally, complete any other connections to V_{DD} or V_{SS} using the 1st metal layer.

At this stage, any connection should be completed. The resulting layout is shown in Fig. 12.

We can now use wider metals for the supply connections (as we usually need to minimize the resistance of these traces) and extend any input and output connections to make it easier to connect our amplifier to other blocks (Fig. 13).

Finally, we need to place pins for any input and output. This is needed to verify that both the schematic and layout have the same I/O ports. You can create some labels by pressing **L** on your keyboard. Type your input and output ports (i.e., “VDD VSS INP INN VB OUT”) in the label field. Then check the option “*use same layer as shape, select purpose pin*”. For a better visibility, you may choose the roman font with a height of 0.5. Then, in your layout editor click on any shape connected to a certain net to place the corresponding I/O port. The final layout is shown in Fig. 13.

5. Design Rule Check (DRC)

After completing the layout, it's essential to verify that it adheres to the design rules. These rules are necessary to enhance the possibility of producing a circuit without errors, given the inevitable variations in the manufacturing process such as mask misalignment and wafer variances. The verification process, known as DRC, checks that the layout complies with the technology's design rules. For example, the metal lines need to be spaced at a specific distance to accommodate any process variations and constraints. It's advisable to run DRC regularly during the design phase to ensure that each modification aligns with all the design rules. In this course, we will use Calibre for DRC verification.

1. To run DRC, click on Calibre > Run > nmDRC. This will open a new Calibre window. In this new window, click on “Run DRC”.
2. Once DRC is complete, a new window will appear (Fig. 14) showing a breakdown of each DRC violation. Each violation category may contain one or more instances, each of which is assigned a unique number. By clicking on an instance, you can view a description of the violated design rule. Additionally, you can highlight a violation instance by right-clicking on it and selecting "Highlight," or by pressing the **H** key on your keyboard. This will cause the Layout Editor window to display flashing white lines indicating the location of the selected violation.

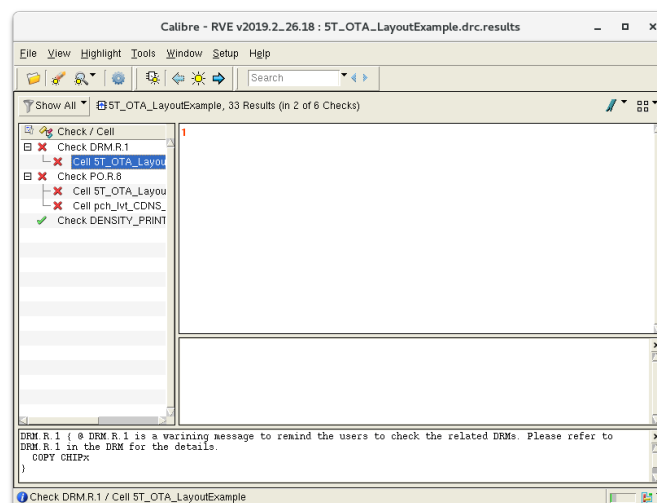


Fig. 14. DRC results

3. If you followed this tutorial carefully you should only see the DRC violations in Fig. 14 (DRM.R.1 and PO.R.8). Fix any other DRC violations. You might want to check the tsmc65 DRM (located in docs/T-N65-CL-DR-001 2 4.pdf) for more details on the violations you are getting.

6. Layout versus Schematic (LVS)

We can now check whether our amplifier layout corresponds to its schematic by running an LVS check. LVS (Layout vs. Schematic) compares the circuit as implemented on the layout against the schematic(s). As designs become more complex, it's easy to make errors such as accidentally connecting two lines or leaving floating nodes, among others. LVS helps to detect such issues by comparing the schematic netlist with the layout's extracted netlist to ensure that they match.

1. Go to Calibre > Run > nmLVS. Wait for a Calibre window to open.
2. In the Calibre window, go to “Inputs”, and under “Netlist” check the “Export from schematic viewer” box. Also, go to “LVS Options” and, under “Supply”, change “Power nets” to VDD and “Ground nets” to VSS (both without a question mark). Also, under “Include”, remove the ./lvs labels.txt entry.
3. Click on “Run LVS”.
4. If other windows pop up click on overwrite and wait until another window (Fig. 15) shows up giving the results on the LVS run.

If your netlists match, then you will see a happy face in the new window, indicating that you have completed the layout successfully. Otherwise, you will see a summary of the LVS problems, in which case you need to correct the layout. Once you are done, close all the Calibre windows.

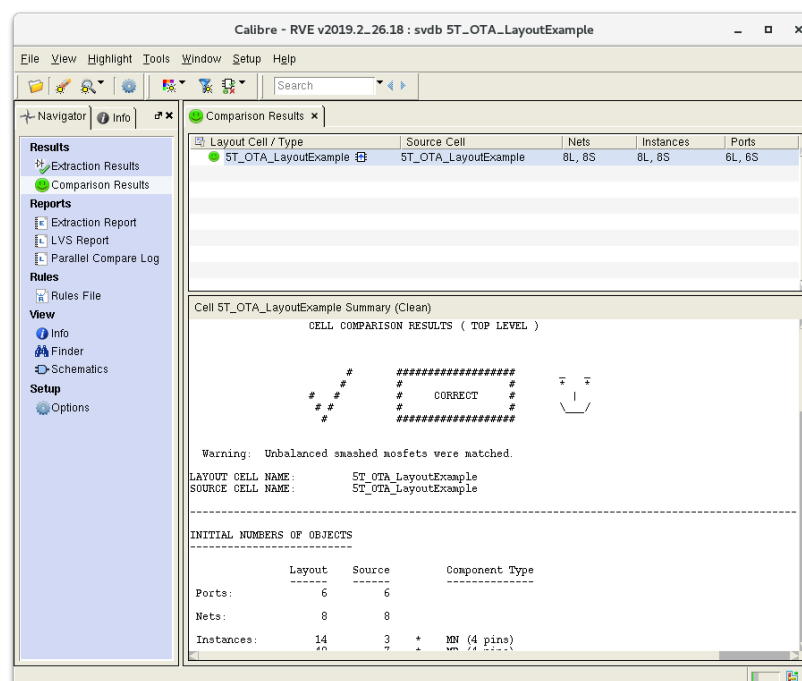


Fig. 15. LVS results

Bonus Exercise

Consider the inverter-based cascode amplifier from the *Sensor Interface Assignment 2* (Fig. 2a). Draw the layout of your amplifier and run the design rule check (DRC) and layout vs. schematic (LVS) check. Fix all LVS and DRC violations, except the DRM.R.1 and PO.R.8 violations. You might want to check the tsmc65 DRM (located in docs/T-N65-CL-DR-001 2 4.pdf) for more details on specific DRC violations that may arise.

List of Shortcuts

1. F2 → Save
2. F3 → Save a copy / Options (when a command is active allows you to access the command options)
3. F5 → Open
4. I → Create instance
5. E → Display options
6. Shift + E → Layout editor options
7. M → Move
8. L → Create label
9. R → Create rectangle
10. Q → Properties (allows you to modify the properties of the selected object)
11. P → Create path
12. A → Align (allows you to align objects)
13. C → Copy
14. O → Create Via
15. K → Ruler
16. S → Stretch (allows you to increase or decrease the size of a selected object)
17. Z → Zoom-in
18. Ctrl + Z → Zoom-in by 2x
19. Shift + Z → Zoom-out by 2x
20. F → Zoom fit
21. Del → delete
22. Ctrl + D → deselect everything
23. Shift + G → Create guard ring
24. Shift + O → Rotate
25. Shift + M → Merge (allows you to merge objects)
26. Shift + K → Remove rulers
27. Shift + J → Flip
28. Shift + C → Chop (allows creating gaps in the selected objects)
29. Shift + F → See lower hierarchical levels
30. Ctrl + F → Hide lower hierarchical levels
31. Shift + P → Create Polygon
32. Ctrl + Shift + X → Create bus
33. Ctrl + Shift + Q → Via up (when P is active)
34. Ctrl + Shift + A → Via down (when P is active)
35. Space → when P is active you can press space to place vias